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MILITARY SPECIFICATION

AMPLIFIER-OSCILLATOR GROUP AN/SIA-114

1. SCOPE

1.1 This specification covers the detailed requirements for an audio amplifier group AN/SIA-114 for general announcing system, circuit 1MC, and intership announcing system, circuit 6MC.

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids, form a part of this specification to the extent specified herein:

SPECIFICATIONS

MILITARY

MIL-C-62 - Capacitors, Fixed, Electrolytic (DC Aluminum, Dry Electrolytic Polarized.)
MIL-I-983 - Interior Communication Equipment, Naval Shipboard, Basic Design Requirements for.
MIL-C-3965 - Capacitors, Fixed, Electrolytic (Tantalum) General Specification for.
MIL-A-21577 - Amplifiers, Audio Frequency; and Amplifier-Control Groups - Shipboard Announcing.

STANDARDS

MILITARY

MIL-STD-200 - Electron Tubes, Selection and Use of.
MIL-STD-701 - Preferrred and Guidance List of Transistors.

(Copies of specifications, standards, drawings, and publications required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Amplifier, AN/SIA-114 shall be in accordance with all applicable paragraphs of Specification MIL-A-21577 and as specified herein. Whenever a requirement of Specification MIL-A-21577 conflicts with a requirement of this specification, the requirement of this specification shall govern.

3.2 Equipment complement shall consist of the following:

(a) One common enclosure, bulkhead mounted, Amplifier control rack, type AN/SIA-114 containing:

(2) Input preamplifier modules type AM-2127/SIA.
(2) Alarm output preamplifier modules type AM-2506/SIA.
(2) Power supply modules type PP-2563/SIA.
(2) Oscillator modules type 0-718/SIA.
(2) Oscillator modules type 0-719/SIA.
(2) Oscillator modules type 0-721/SIA.
(2) Oscillator modules type 0-722/SIA.
(2) Oscillator modules type 0-724/SIA.
All necessary switches, meters, relays, wiring, terminals, and other components required.

(b) One common enclosure, deck mounted, power amplifier assembly type AM-2316/SIA containing two 500 watt audio frequency amplifier assemblies type AM-2128/SIA.

3.3 Modular construction techniques with quick disconnect features shall be employed in the design of these assemblies to permit easy removal of the modules for servicing.

3.4 All modules shall be mounted in a vertical position to facilitate the unrestricted flow of air between modules for ventilation purposes.

3.5 Capacitors. - Tantalum electrolytic capacitors conforming to Specification MIL-C-3965 shall be used in preamplifiers and oscillator assemblies. Plug in type electrolytic capacitors may be used in power supply modules type PP-2563/SIA. These capacitors shall be in accordance with Specification MIL-C-62.

3.6 Wiring. - Wiring employed shall have insulation type T4 as shown in applicable table of Specification MIL-I-983.

3.7 Transistors. - Transistors used shall be in accordance with Standard MIL-STD-701.

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3.8 Diodes. - Metallic diodes used shall be in accordance with Standard MIL-STD-200.

3.9 Forced air cooling. - In addition to the requirements specified in Specification MIL-A-21577, facilities for forced air cooling will be required for the power amplifiers. Blower motors shall be Rotron type KS551, or equal. Blower motors shall be individually fused.

3.10 Time constant. - Under the conditions specified in the overload limiter characteristic requirements of Specification MIL-A-21577, the preamplifier output voltage shall return to the value specified in the overload limiter characteristic requirements within 0.025 second after the sudden increase in input signal voltage above reference level.

3.11 Relays. - All microphone input relays, amplifier input relays, alarm input relays, and alarm output relays shall be miniature, hermetically sealed d.c. relays with bifurcated contacts. All control relays shall be miniature, hermetically sealed d. c. relays.

3.12 Test facilities. - A test meter and switching facilities shall be included in each power amplifier and shall provide a ready means for indicating the operating conditions of each vacuum tube stage, and the output signal level. A common voltmeter with switching facilities shall be provided on the control rack to determine proper power supply voltage for amplifier-oscillators, and preamplifiers. The meter calibration and circuit component values shall be such that the operating condition of the amplifiers or amplifier-oscillators can be determined directly from the meter without reference to tables or charts.

3.13 Output power. - The rated power for each of the two power amplifier assemblies shall be 500 watts, minimum rating.

3.14 Interstage input circuit, input impedance and input level will not be required.

3.15 Each of the amplifier-oscillator assemblies shall be capable of generating the following alarm signals:

Signal number	Signal function
1	Collision alarm
4	Chemical alarm
5	General alarm
7	Unassigned
2	Flight crash alarm

3.16 Generated signals. - Alarm signals numbers 3, 6, and 8 will not be required. Electronic timing shall be provided on circuits where timing is required.

3.17 Output level. - The amplifier-oscillator shall be capable of maintaining an output level of not less than 10 volts into a 600 ohm resistive load for amplifier input. This voltage shall also be used as a test signal source as specified in Specification MIL-A-21577.

3.18 Adjustment of output level. - A screwdriver type adjustment with shaft-locking nut shall be provided in the amplifier-oscillator or in combination with the associated control circuits to permit independent adjustment of the amplifier-oscillator output level for the alarm signal into the amplifier input.

3.19 Input facilities. - Amplifier, AN/SIA-114 shall have the following input facilities:

Microphone station number	Microphone station type	Circuits controlled
1	IC/MSB-2	1MC-6MC
2	IC/MSB-2	1MC
3	IC/MSB-2	1MC
4	IC/MSB-2	1MC
5	IC/MSB-2	1MC
6	IC/MSB-2	1MC

3.20 Circuit attenuation. - Operation of the 1MC system from any microphone station shall energize a circuit that is capable of attenuating circuit SE. This circuit shall be capable of making and breaking a resistive load of 7 amperes at 115 volts 60 cycle single phase.

3.21 Local loudspeaker cutout circuit. - A cutout circuit shall be included for each microphone station to automatically disconnect local loudspeakers when the microphone is in use. The cutout circuit shall consist of a normally closed relay which shall operate to open the audio circuit supplying the local loudspeakers when the microphone press-to-talk switch is operated. These relays shall be installed in the amplifier control rack. Six local cutout circuits are required.

3.22 Output facilities. - Amplifier, AN/SIA-114 shall have the following output facilities:

Loudspeaker group number	Loudspeaker group designation	Circuit
1	Officers	1MC
2	Topside	1MC
3	Crew	1MC
4	Engineering spaces	1MC
5	Bull horn	6MC
6	Spare	1MC

3.23 Spare switches. - One spare amplifier output switch shall be provided.

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3.24 Input and output metering and switching shall apply as specified in 3.25.

3.25 Combined metering. - A common volume indicator meter shall be provided for input and output metering. This meter shall indicate zero DB at rated input and output power.

3.26 Test input level control. - Test input level control will not be required.

3.27 Microphone press-to-talk switches. - Reduction in output of circuit SE, ships entertainment system following operation of a press-to-talk switch, shall not apply to the 6MC system.

3.28 Alarm signals shall not be provided for circuit 6MC.

3.29 Visual alarm. - Operation of alarm signals numbers 1, 4, 5 and 8 shall initiate operation of the visual alarm signal.

3.30 Circuit SE muting. - Operation of any alarm contact shall energize a circuit that is capable of muting circuit SE. This circuit shall be capable of making and breaking a resistive load of 7 amperes at 115 volts 60 cycle single phase.

3.31 Alarm signal priorities. - The order of priority for alarm signals shall be as follows:

(a) Collision alarm.
(b) Chemical attack alarm.
(c) General alarm.
(d) Unassigned.
(e) Flight crash alarm.

The oscillator modules shall be provided with a device to prevent unauthorized or accidental changes in alarm signal priorities.

4. QUALITY ASSURANCE PROVISIONS

4.1 Sampling, examination and tests shall be as specified in Specification MIL-A-21577.

5. PREPARATION FOR DELIVERY

5.1 Preparation for delivery shall be as specified in Specification MIL-A-21577.

6. NOTES

6.1 The notes specified in Specification MIL-A-21577 are applicable to this specification.

Notice. - When Government drawings, specifications, or other data are used for any purpose other than in connection with a definitely related Government procurement operation, the United States Government thereby incurs no responsibility nor any obligation whatsoever; and the fact that the Government may have formulated, furnished, or in any way supplied the said drawings, specifications, or other data, is not to be regarded by implication or otherwise as in any manner licensing the holder or any other person or corporation, or conveying any rights or permission to manufacture, use, or sell any patented invention that may in any way be related thereto.

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